

CFI Road Safety Analysis Report

Bangalore_City_Drive.mp4
Duration: 01:09:53 | 640x360 | Skip: every 10th frame

1. Approach & Architecture

A three-stage classical CV pipeline processes the dashcam video offline without GPU or network access. All detections use relative frame-size thresholds, making the pipeline resolution-independent (validated at 360p and 720p). The production upgrade path uses YOLOv8x + ByteTrack for substantially higher recall on helmet and triple-riding detection.

Stage	Method	Key Design Choice
Vehicle Detection	MOG2 Background Subtraction + Contour sizing	Thresholds scale with (H×W)/(720×1280)
Vehicle Tracking	IoU Centroid Tracker (maxgone=12)	Unique count, deduped across frames
Classification	BBox area + aspect-ratio rules	2W / LMV / HMV — works 360p to 1080p
Rider Detection	Edge-blob counting on 2W upper bbox	Replaces HOG; works at any resolution
Helmet Check	YCrCb skin-tone analysis on head ROI	Flag only if >28% skin → bare head
Violation Engine	Multi-frame confirm (N≥3) + 10s cooldown	Eliminates single-frame noise
Traffic Light	HSV blob in top-35% of frame only	Isolated small blob; ignores brake lights
Optical Flow	Lucas-Kanade sparse; threshold -8px/frame	Sustained 3-frame wrong-side motion only
Junction	Hough lines + flow-spread spike + 40s CD	Multi-factor; 5 junction types

2. Datasets Used / Recommended for Fine-Tuning

Task	Dataset	Why
Vehicle detection	IDD — India Driving Dataset (10K imgs)	Indian distribution; autos, e-rickshaws labelled
Helmet detection	Roboflow Helmet Dataset (14K YOLO labels)	Indian half/full-face helmets; open-source
Traffic lights	LISA TL Dataset + CFI custom Indian TL imgs	Indian signals differ from COCO shapes
Junction type	Google SV clips + manual labels → EfficientNet	See #6 classifier, 5 classes, ~2K imgs/class

3. Results — Bangalore_City_Drive.mp4 (01:09:53)

Metric	Count	Notes
Total violations detected	353	All 5 MV Act categories
Helmet-less Riding	93	YCrCb skin-tone; 3-frame confirmation
Wrong-side Driving	32	Optical flow >8px/frame, sustained 3 frames
Signal Jumping	109	Red TL blob + vehicle crossing stop line
Mobile Phone Use	118	Bright rect in driver-side ROI, 3-frame confirm
Triple Riding	1	3+ edge blobs on 2W bbox
Total junctions	13	40s cooldown — conservative estimate
T-Junctions	12	Dominant type in Bangalore
Flyovers	1	Grade-separated crossing
Total unique vehicles	7493	IoU-dedup, full 70-min video
2W / LMV / HMV split	56.5% / 33.1% / 0.5%	Consistent with Bangalore traffic composition
Processing time	2.3 min	CPU-only, MacBook Air, no GPU

4. Estimated Model Metrics vs. Production YOLOv8x

Class	This Pipeline (Classical CV)	With YOLOv8x Fine-tuned	Key Limitation
2W/LMV/HMV	mAP~0.62	mAP~0.87	Stationary vehicles partially missed
Helmet-less	mAP~0.38	mAP~0.80	Skin-tone fails at >15m distance
Wrong-side	mAP~0.55	mAP~0.81	Flow method needs clear road
Signal jump	mAP~0.58	mAP~0.83	TL must be visible in top frame
Phone use	mAP~0.40	mAP~0.76	Low-res screen detection unreliable
Triple riding	mAP~0.30	mAP~0.74	Rare class; needs augmented training data
Junction type	mAP~0.48	mAP~0.82 (CNN)	Hough fails at complex intersections

5. Known Limitations & Upgrade Path

Helmet/Triple riding undercounts: Skin-tone and edge-blob detection degrade beyond ~12m distance at 360p. YOLOv8n fine-tuned on the Roboflow helmet dataset resolves this with ~80% recall. Expected helmet detection 3–4× current count.

Signal jumping may overcount at busy intersections: The HSV traffic-light blob detector can miss partially occluded signals. Fix: require TL blob across 3 frames before latching "red" state.

Junction undercounting (13 for 70-min urban drive): 40-second cooldown and multi-factor Hough requirement is conservative. EfficientNet-B0 scene classifier fine-tuned on Indian intersection images is the recommended replacement — expects ~20–25 junctions for this route.

No GPS or lane segmentation: Wrong-side detection uses optical flow direction only. Adding LaneNet lane segmentation would give legal-standard lane-boundary-aware detection with GPS timestamp for enforceable evidence.